GATE PYQs

GATE 2024

```
def fx():
    a = input()
    if 'n' not in a:
        fx()
        print(a)
fx()
```

Assume that the inputs to the function are given in the following order: 1, 2, 3, 4, n

What is the output of the function?

In [ ]:
```
def fx():
    a = input()
    if 'n' not in a:
        fx()
        print(a)
fx()
```

```
4
3
2
1
```

GATE 2023

```
def f1():
    return 1

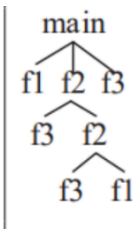
def f3() -> int:
    return 5

def f2(x: int) -> int:
    f3()
    if x == 1:
        return f1()
    else:
        return x * f2(x-1)

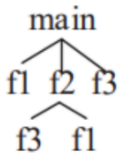
f1()
f2(2)
f3()
```

Which one of the following represents the activation tree corresponding to the main code?

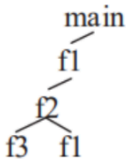
(A)



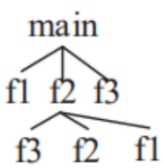
(B)



(C)



(D)



opt. a

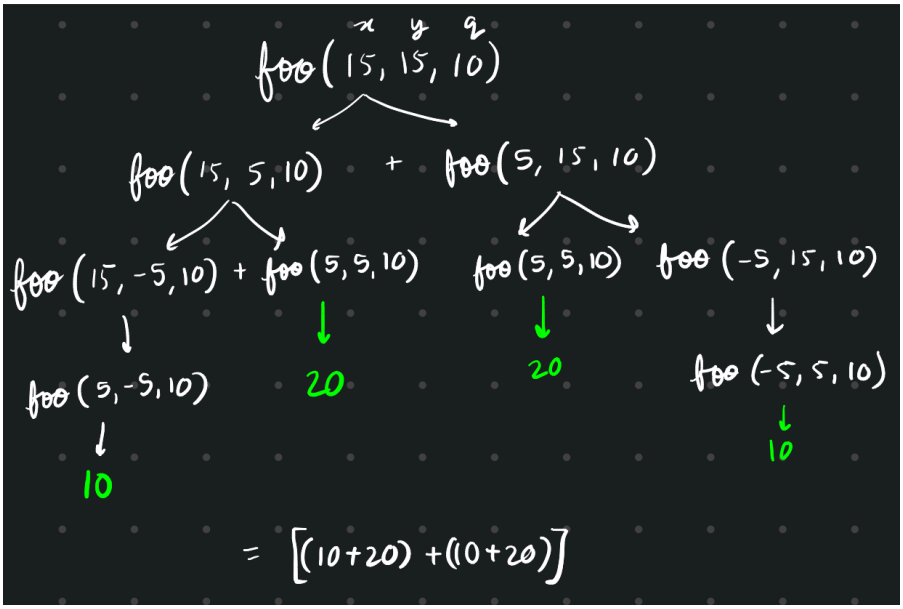
GATE 2021

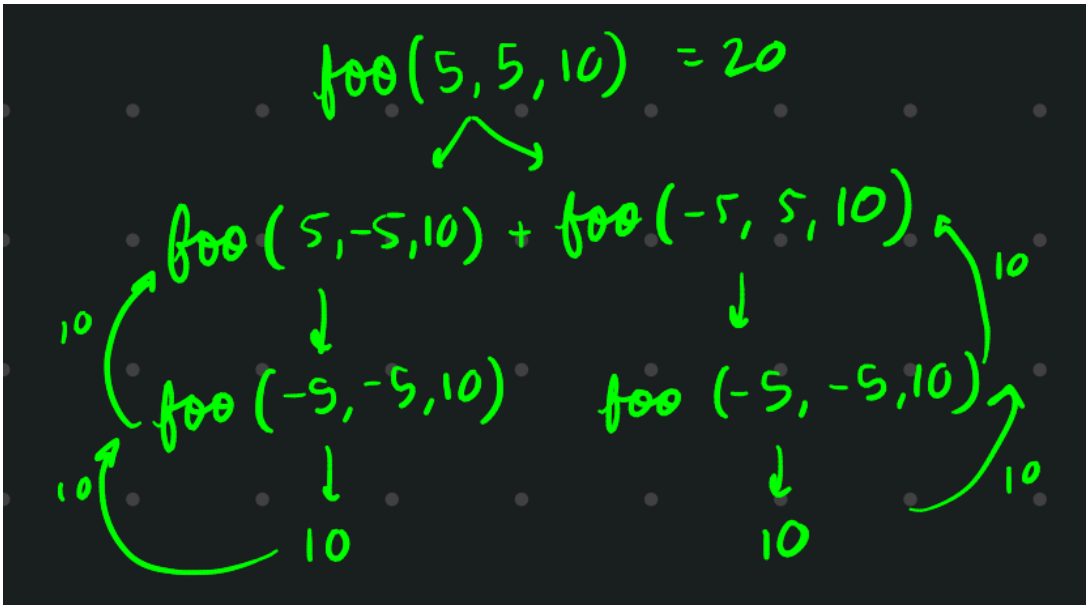
```
def foo(x, y, q):
    if x<=0 and y<=0:
        return q
    if x<=0:
        return foo(x, y-q, q)
    if y<=0:
        return foo(x-q, y, q)
    return foo(x, y-q, q) + foo(x-q, y, q)

r = foo(15, 15, 10)
print(r)
```

What is the output of the code?

a. 20 b. 220 c. 160 d. 100 e. 60 f. None of the above g. Error





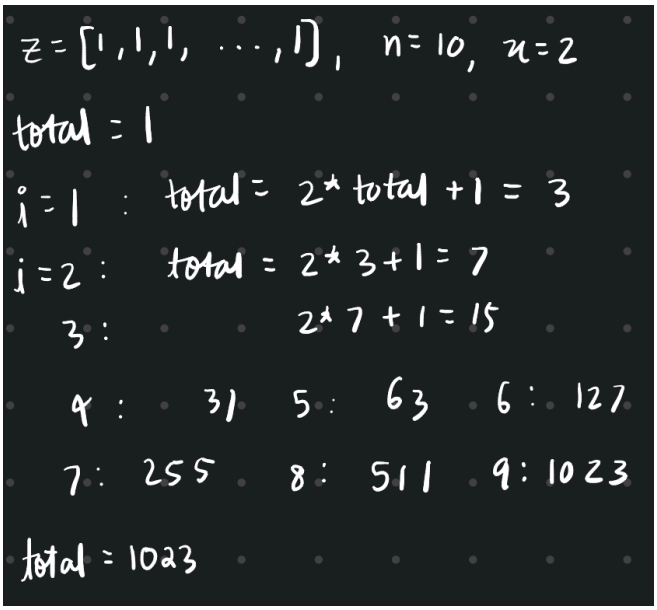
opt. e. 60

GATE 2021

```
def simple_function(Y, n, x):
    total = Y[0]
    for i in range(1, n):
        total = x*total + y[i]
    return total
```

Let Z be an array of 10 elements with Z[i] = 1 for all i such that 0 <= i <= 9. What is the output of the following code snippet?

```
simple_function(Z, 10, 2)
```



GATE 2020

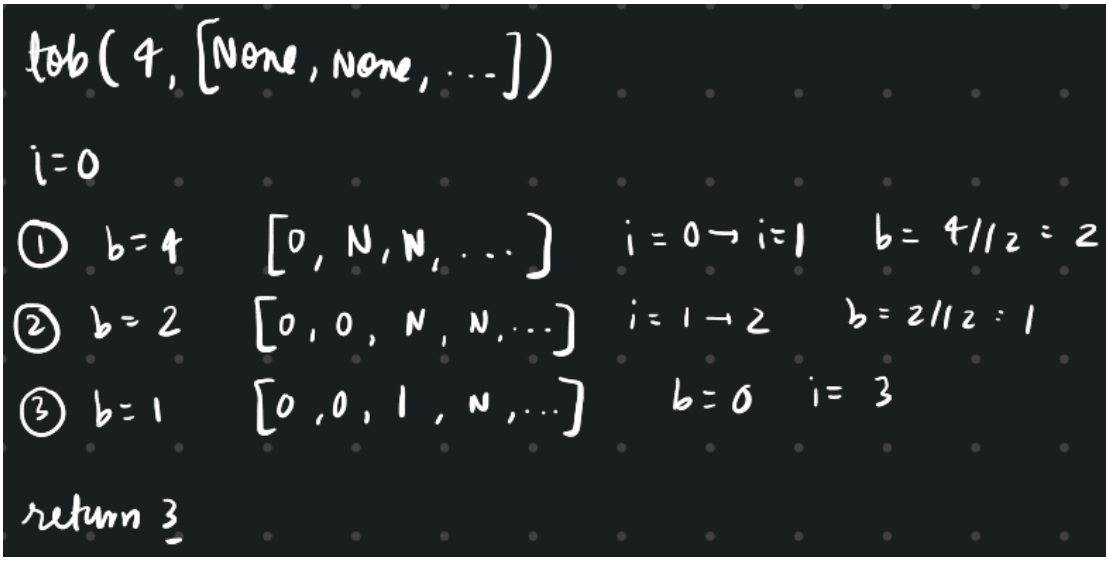
```
def tob(b, arr):
    i = 0
    while b > 0:
        if b%2:
            arr[i] = 1
        else:
            arr[i] = 0
        b = b//2
        i += 1
    return i

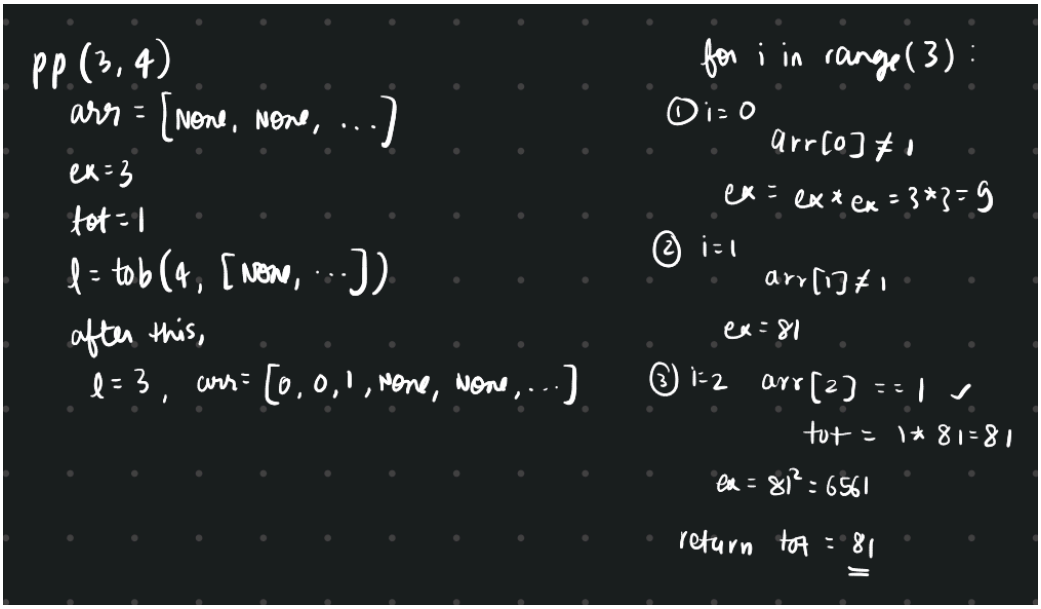
def pp(a, b):
    arr = [None] * 20
    ex = a
    tot = 1
    l = tob(b, arr)
    for i in range(l):
        if arr[i] == 1:
            tot = tot * ex
        ex = ex * ex
```

```
    return tot

print(pp(3, 4))
```

What is the output of the code?



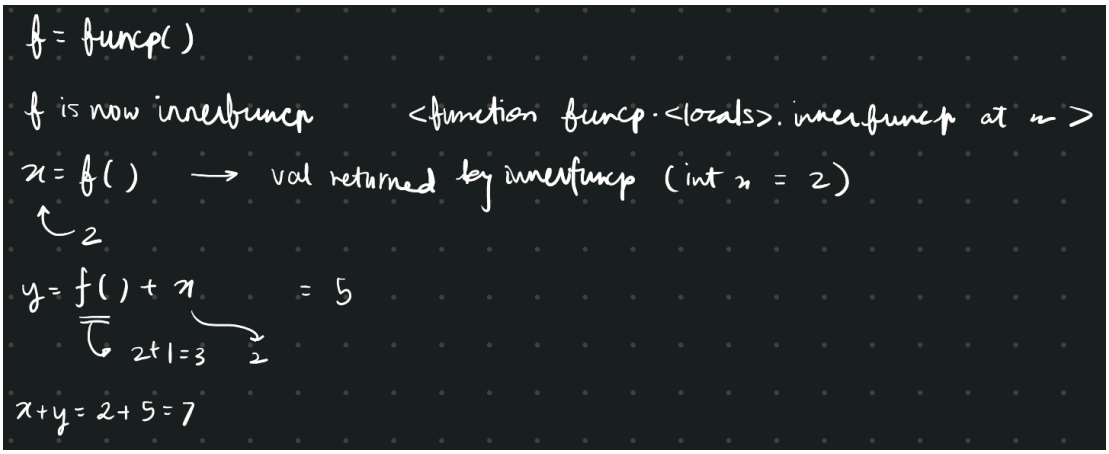


GATE 2023

```
def funcp():
    x = 1
    def innerfuncp():
        nonlocal x
        x += 1
        return x
    return innerfuncp

f = funcp()
x = f()
y = f() + x
print(x+y)
```

What is the output of the code?

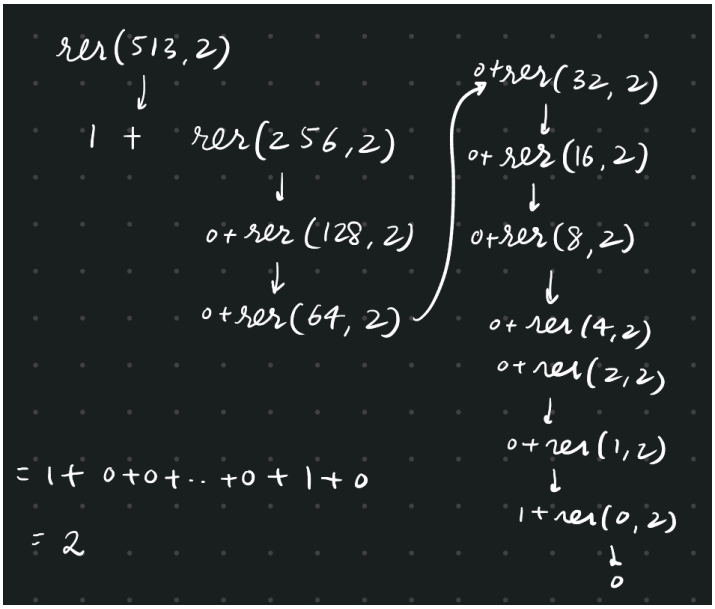


ISRO 2020

```
def rer(n, r):
    if n>0:
        return(n%r + rer(n//r, r))
    else:
        return 0

print(rer(513, 2))
```

What is the output of the code?



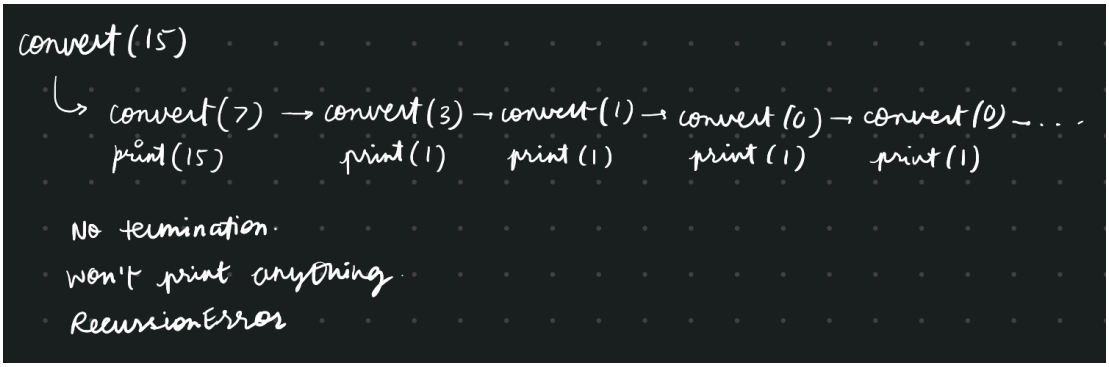
GATE 2019

```
def convert(n):
    if n < 0:
        print(n)
    else:
        convert(n//2)
        print(n)
```

Which of the following will happen when convert() is called with any positive integer as argument?

a. It will print the binary representation of the number.

b. It will print the binary representation of the number in reverse order.
c. It will print the binary representation of the number but will not terminate.
d. None of the above.



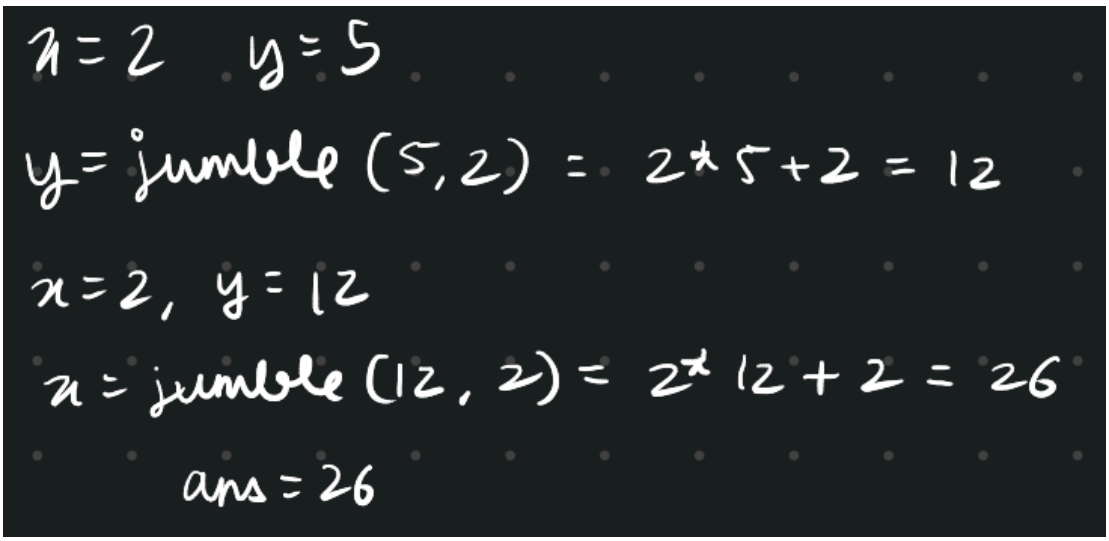
GATE 2019

```
def jumble(x, y):
    x = 2*x+y
    return x

x=2; y=5
y = jumble(y, x)
x = jumble(y, x)
print(x)
```

What is the output of the code?

a. 13 b. 26 c. 2 d. 5

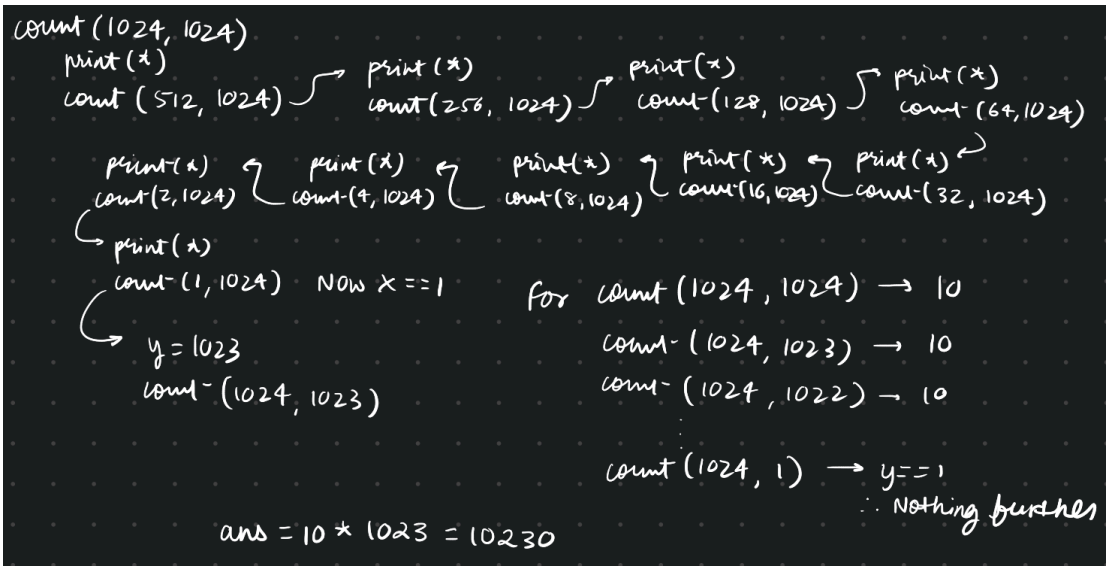


GATE 2018

```
def count(x, y):
    if y != 1:
        if x != 1:
            print('*')
            count(x//2, y)
        else:
            y -= 1
            count(1024, y)
```

`count(1024, 1024)`

The number of times the print statement is executed when count(1024, 1024) is called is _____.



GATE 2017

```
def foo(val):
    x = 0
    while val > 0:
        x = x + foo(val)
        val -= 1
    return val

def bar(val):
    x = 0
    while val > 0:
        x = x + bar(val-1)
    return val
```

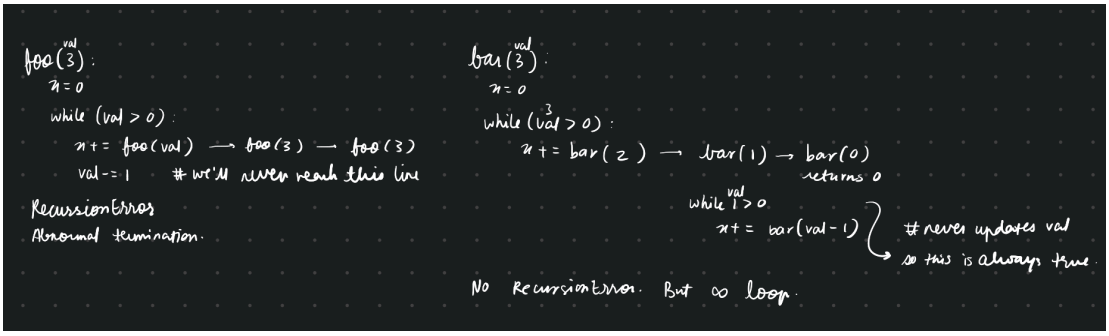
Invocations of foo(3) and bar(3) will result in _____.

a. 6 and 6 respectively.

b. infinite loop and abnormal termination respectively.

c. abnormal termination and infinite loop respectively.

d. both abnormal terminations.



GATE 2017

```
def fun1(n):
    if n == 0:
        return
    fun2(n-2)
    print(n)

def fun2(n):
    if n == 0:
        return
    print(n)
    n += 1
    fun1(n)
    print(n)
```

The output printed when `fun1(5)` is called is _____.

a. 5 3 4 2 3 1 2 0 1 1 2 2 3 3

b. 5 3 4 2 3 1 2 2 2 3 3 4 4 5

c. 5 3 4 2 3 1 2 2 1 3 2 4 3 5

d. 5 3 4 2 3 1 2 0 2 1 3 2 4 3